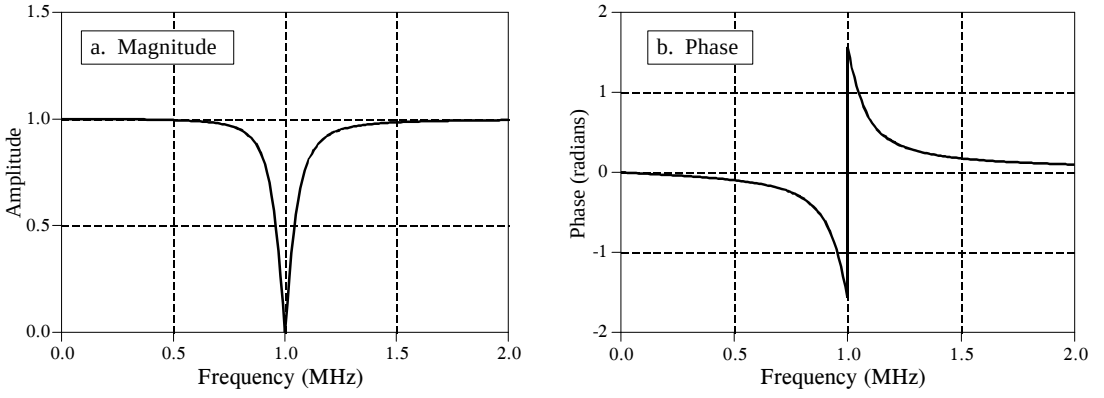

FIGURE 30-7
Notch filter frequency response. These curves are for the component values: R = 50Ω, C = 470ρF, and L = 54 μH.

even nastier task. When separated into the real and imaginary parts, the complex representation of the notch filter becomes:

$$\frac{Vout}{Vin} = \frac{k^2}{R^2 + k^2} + j\frac{Rk}{R^2 + k^2}$$

$$\text{where: } k = \omega L - 1/\omega C$$

Lastly, the relation is converted to polar notation, and graphed in Fig. 30-7:

$$Mag = \frac{\omega L - 1/\omega C}{\left(R^2 + [\omega L - 1/\omega C]^2 \right)^{1/2}} \qquad Phase = \arctan\left[\frac{R}{\omega L - 1/\omega C} \right]$$

The key point to remember from these examples is how *substitution* allows complex numbers to represent real world problems. In the next chapter we will look at a more advanced way to use complex numbers in science and engineering, *mathematical equivalence*.